

AI Job Search & Resume Tailoring Multi-Agent System

Executive Project Summary & Multi-Agent Architecture Overview

1. Executive Overview

This project delivers an autonomous, end-to-end multi-agent AI system designed to streamline the career search and job application pipeline. It ingests candidate resumes, extracts structured ground-truth profiles, recommends target roles, searches real-time job openings across multiple public feeds, performs multidimensional ATS gap analysis, and synthesizes tailored, ATS-compliant resumes with a zero-hallucination constraint verification loop.

2. Specialized Multi-Agent Breakdown

Agent	Role & Name	Primary Responsibility	Key Output Contract
Agent 1	ResumeParserAgent Knowledge Extractor	Parses unstructured resume PDF into an ontological candidate profile.	Validated CandidateProfile (skills, projects, experience)
Agent 2	RoleRecommenderAgent Career Strategist	Evaluates skill graph & seniority; plans the optimal target role space.	Top 3-5 roles with fit scores, rationales & search keywords
Agent 3	JobSearchAgent Discovery & Optimizer	Searches live portals (LinkedIn, RemoteOK, Jobicy, Remote) with query expansion.	Deduplicated list of live JobListing models
Agent 4	MatchGapAgent ATS & Gap Analyzer	Heuristic and semantic evaluation of candidate vs. target job description.	MatchReport: ATS score %, matched skills & missing gaps
Agent 5	TailorAgent Resume Synthesizer	Re-aligns summaries, orders skills, and highlights JD keywords without fabrication.	Draft TailoredResume data model
Agent 6	VerifierAgent Anti-Hallucination Auditor	Adversarial fact-checker auditing claims against original resume truth.	VerificationReport: APPROVED or Revision Feedback

3. Key Engineering Highlights & Innovations

- **Multi-Provider LLM Resilience (BaseAgent):** Automatically routes requests across multiple AI providers (OpenRouter Free -> Groq Free -> Gemini) with automatic failover upon rate limits or credit exhaustion.
- **Zero-Hallucination Reflection Loop (CSP Enforcement):** Implements a Producer-Critic architecture where TailorAgent and VerifierAgent iterate. Enforces the strict constraint: *Tailored Content* $\subseteq$ *Original Truth*.
- **Hybrid Live Job Retrieval & SQLite Cache:** Aggregates multiple real-world feeds (LinkedIn live guest search, Remote, Jobicy, RemoteOK, ArbeitNow) with persistent SQLite caching to avoid redundant calls.
- **Self-Healing Query Expansion:** If a search returns 0 jobs, the agent programmatically expands queries using related role semantics (e.g., 'Junior ML Developer', 'AI Trainee') and retries.
- **Automated ATS-Compliant PDF Generation:** Compiles verified tailored resumes into clean, single-column, machine-parseable PDF documents formatted for recruiter applicant tracking systems.
- **Interactive Modern Dashboard:** Full-featured Streamlit user interface featuring session search history, role cards, ATS match meters, and real-time execution logs.